Adsorption of Sodium Alkyl Sulfate Homologues at the Air/Solution Interface

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Experimental results are presented on the adsorption of sodium alkyl sulfate homologues ($n_C = 8-14$) at the air/solution interface. The adsorption isotherms calculated from equilibrium surface-tension vs concentration data and the critical micelle concentration change regularly with the length of the alkyl chain; the odd/even effect was not observed. The isotherms were analyzed using a model-independent approach. The analysis indicates that the total driving force of adsorption reaches a plateau value and becomes constant in the function of the adsorbed amount in the case of each homologue. With the use of different electrostatic models, it was demonstrated that this behavior is consistent with a saturation-type hydrophobic driving-force contribution, which can be interpreted by the development of a liquidlike alkane environment in the adsorbed layer above a "critical" adsorbed amount.

Introduction

The interfacial properties of the surfactant solutions determine and control the performance of many practical applications.[1] This explains the widespread interest in surfactant adsorption studies and the fact that this phenomenon is still a subject of intensive experimental and theoretical investigations.[2]

The generally accepted physical picture of surfactant adsorption is that the adsorbed surfactant molecules form an oriented monolayer at the solution interface. The major driving force of surfactant adsorption is the hydrophobic driving force, which is attributed to the removal of the hydrocarbon chains from the aqueous environment. However, the detailed physical picture is far more complicated and should explain the following experimental observations of (1) the shape of the adsorption isotherms, (2) the odd/even effect of the alkyl chain length, (3) the dependence of the adsorption on the concentration of added electrolytes, and (4) the effect of the chemical type of the surfactant counterions.

The aim of the present paper is to study the role of the alkyl chain length in the adsorption of the sodium alkylsulfate homologues at the free aqueous interface. The investigation addresses the existence of the odd/even effect and the model-independent determination of the total adsorption driving force as well as its division into an electrostatic and a hydrophobic contribution.

Theoretical Background

The simplest approach used to describe the adsorption of surfactant molecules is the Langmuir model. The model assumes the adsorption of the surfactant molecules on N_0 adsorption sites with constant driving force, thus neglecting the interaction between the molecules within the adsorption layer. To improve this approach, several different equations of state have been developed in the literature. The interested reader can find a comprehensive review of them, e.g., in ref 3. In surfactant adsorption studies one of the most extensively used extensions of the Langmuir model is the Frumkin model,[4] which takes into account the lateral interaction of the adsorbed surfactant molecules by means of the Bragg−Williams lattice model giving rise to the linear dependence of the hydrophobic driving force on the surface coverage. However, it should be noted that the surface equations of state are derived assuming a monolayer of disklike molecules. This assumption may be reasonable for a diluted surfactant monolayer when the adsorbed molecules are far from each other, but one can expect its breakdown when the surfactant alkyl chains start to overlap in the adsorption layer. This theoretical expectation is in line with the thorough analysis of surface-tension data, which indicated that by using either the Langmuir or the Frumkin model only the lower or the upper part of the surface-tension isotherms could be described, but a satisfactory fit in the entire concentration range could not be achieved.[5,6] Furthermore, a number of experimental studies of the high-frequency surface dilational elastic modulus indicated that the experimental values are much smaller than the values calculated from the Langmuir or Frumkin isotherms.[7−9] To overcome these discrepancies, a few theoretical approaches were suggested, which either assumed that the surfactant molecules can adsorb in two distinct states in the adsorption layer and with increasing surface coverage the reorientation of the molecules takes place[10−13] or assumed the intrinsic compressibility of the interfacial layer.[14−17]

In the case of ionic surfactants, the adsorption isotherms have been described at different levels of approximations in the literature. The simplest approach is the application of the ''pseudo-nonionic" models. In this case, an adsorption isotherm derived for nonionic surfactants is used for the fitting of the experimental data. These models give limited physical insight due to the neglected electrostatic interactions and cannot interpret any of the specific characteristics of the experimental isotherms.[2]

The second group of the adsorption isotherms of ionic surfactants incorporates an electrostatic contribution in the adsorption free energy by means of the Gouy−Chapman theory.[18] One of the underlying assumptions of these models is that the real, continuous spatial charge-density distribution of

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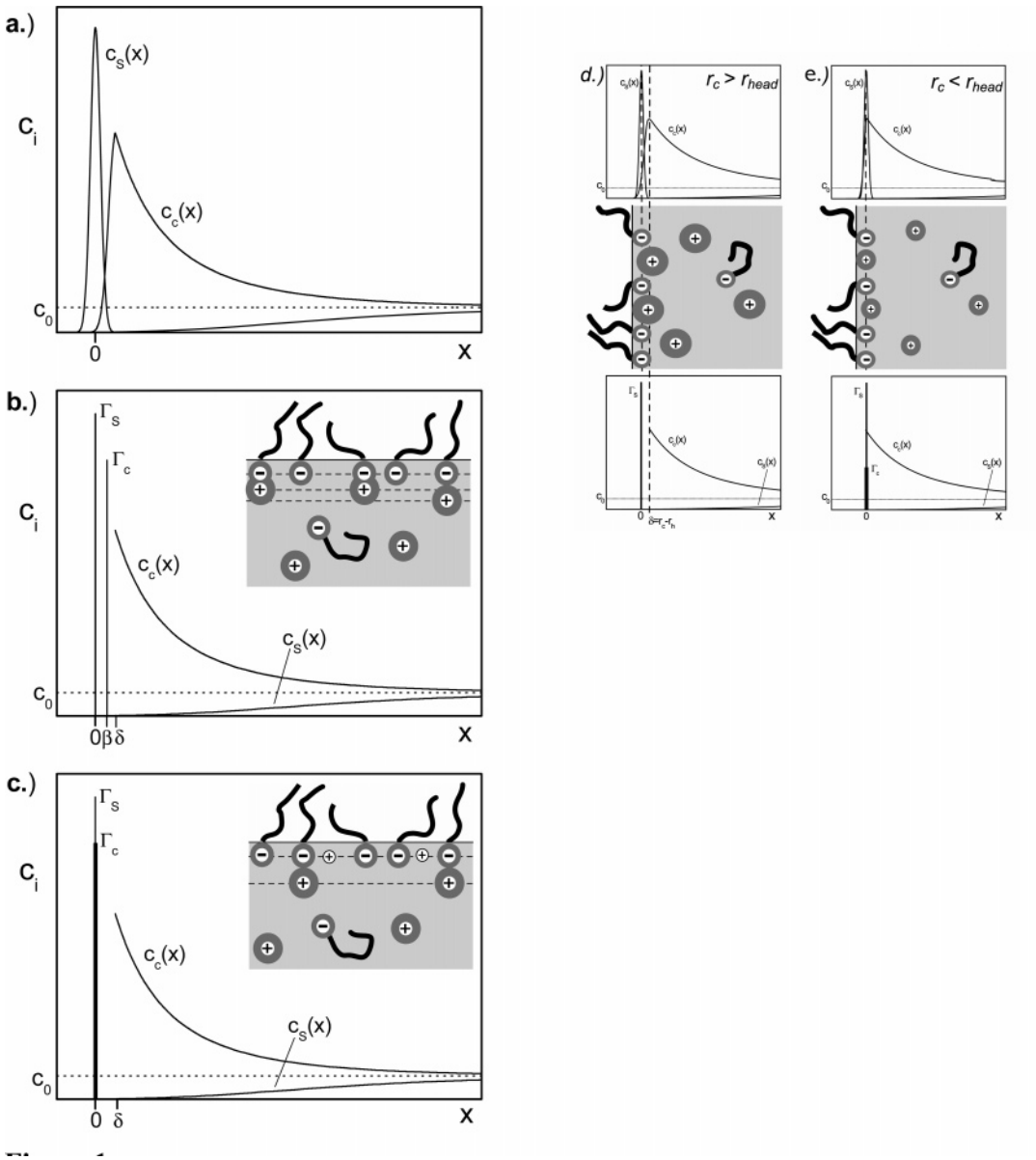


Figure 1. Schematic representation of the continuous spatial charge-density distribution of the double layer (a and upper panels of d and e) and its modeling by Kalinin and Radke[21] (b), Warszynski et al.[28] (c), and Gilanyi et al.[30] (d and e). The insets of the figures and the middle panel of d and e depict the models of the Stern layer. The dashed lines denote the plane of the surfactant headgroups, the plane of the counterion binding, and the plane of the closest approach.

the electric double layer (see Figure 1) can be divided into a compact and a diffuse part. The adsorption models developed for ionic surfactant adsorption differ from each other in the geometrical model used to accomplish this division. First Davies[19] then Borwankar and Wasan[20] used the Gouy−Chapman theory in its original form. They assumed that the fully dissociated, adsorbed surfactant molecules form a uniformly charged monolayer and the pointlike counterions build up the diffuse part of the double layer. These models can successfully describe the experimental surface-tension isotherms; however, on one hand they cannot interpret the effect of counterions on the surfactant adsorption; on the other hand they predict too high electric potentials at the Helmholtz plain, which gives rise to physically unrealistic counterion concentration (∼30 M) in the close vicinity of the surfactant monolayer.

To overcome this problem, different counterion binding models were proposed in the literature. First Kalinin and Radke[21] assumed that counterions can bind to the adsorbed surfactant molecules in an equilibrium chemical reaction. They described the interface by a triple-layer model (Figure 1b), originally proposed by Grahame[22] for a mercury/electrolyte interface. The first plane contains the adsorbed surfactant molecules. The second one (inner Helmholtz plane) contains the bound counterions, while the third plane (outer Helmholtz plane) represents the closest approach of the hydrated counterions of the diffuse layer to the adsorbed surfactant ions. The proposed model was successfully fitted to surface-tension data measured for aqueous sodium dodecyl sulfate (SDS) solutions at air/solution and oil/solution interfaces. However, it should be noted that the model assumes strong chemical interactions between the adsorbed surfactant headgroups and the counterions, which is a realistic assumption for weak electrolyte-type surfactants (e.g., organic fatty acids) but seems to be unnatural for strong electrolyte-type surfactants (e.g., alkali alkyl sulfates) that do not have the ability to complex (to form covalent bonds) with the counterions. For example, in the case of SDS, the equilibrium constant of counterion binding was determined as 61.3 dm^3/mol. Taking into account that the adsorption of the surfactant ions does not

have an effect on the electron structure of the surfactant (physisorption), the same equilibrium constant must characterize the counterion binding to the surfactant in the bulk, which should give rise to ~30% ion pair formation at 8 mM SDS concentration in the bulk (at this concentration SDS is fully dissociated).

It was suggested a long time ago that since there may be considerable space between the headgroups of the adsorbed surfactant molecules, the counterions can penetrate into the plane of headgroups.[23−27] Warszynski et al.[28] utilized this assumption in a model that assumes that the counterions can adsorb in the same plane where the primary charges (the surfactant headgroups) are situated, while a second plane (the outer Helmholtz plane) defines the closest approach of the hydrated counterions of the diffuse layer (Figure 1c). Since in the framework of this model the counterions bind to empty surface sites, to avoid the binding of the counterions to the bear surface, the counterion binding must be electrostatically driven. To get great enough driving force for the counterion binding, a thick (4 Å), charge-free Stern layer was defined a priori between the adsorption monolayer and the Helmholtz plane with a low dielectric constant (30). However, it should be noted that the model prediction that the counterions can enter the adsorption plane mostly without losing their hydration shell but are expelled from a thick Stern layer seems to be inconsistent. On the other hand, the model allows a similar quality fit to experimental surface-tension data as the model of Kalinin and Radke; furthermore, it can interpret the counterion effect on the adsorption.[29]

Recently, Gilanyi et al.[30] proposed an alternative counterion penetration model. They assumed that the closest approach of the mobile counterions to the water/air interface is limited by their hydrated size (Figure 1d,e). As a consequence, if the hydrated counterion size is smaller than that of the surfactant headgroup, the counterions can enter the adsorption plane and reduce the surface charge density (Figure 1e). On the other hand, if the hydrated counterions are larger than the surfactant headgroups, then a charge-free Stern layer is formed with a thickness equal to the difference of the hydrated counterion and headgroup size (Figure 1d). The model successfully describes experimental surface-tension data and interprets the counterion effect but in the case of "large" counterions predicts too high electrostatic potential and consequently physically unrealistic counterion concentration at the Helmholtz plane.

To describe the nonelectrostatic part of the driving force in the electrostatic models, usually the Langmuir[19] or the Frumkin[20] model is used. However, recently a few alternative approaches have also been utilized. Mulqueen and Blankschtein[31] used a one-parameter, molecular thermodynamic theory developed by Nikas et al.[32] for nonionic surfactants. In this model, the surfactant monolayer is treated as a two-dimensional gas of hard disks interacting via van der Waals interactions. Gilanyi et al.,[30] based on the results of recent theoretical[33] and experimental investigations,[34,35] used the underlying physical picture that above a certain surface coverage the overlapping alkyl chains of the adsorbed surfactant molecules form a thin liquidlike alkane layer at the interface. This physical picture implies a driving-force function that quickly approaches a limiting value since considerable changes in the lateral interaction among the alkyl chains can be expected only at low surface coverage (before the formation of the liquidlike layer). They approximated the driving-force function with two linear regions. At low surface coverage they used a Frumkin-like driving-force function, while after the formation of the liquidlike alkane layer constant hydrophobic driving force was used. Very recently, Stubenrauch et al.[36] used the so-called intrinsic molecular compressibility model of Fainerman et al.[17] for the description of the adsorption of alkyl trimethylammonium bromides. This model assumes the decrease of the molar area of the surfactant with increasing surface pressure due to the reorientation of the alkyl chains.

Finally, it should also be noted that Lunkenheimer et al.[37] investigated the adsorption of sodium alkyl sulfate homologues (from C_7 to C_{14}) by surface-tension measurements on surface chemically purified solutions. It was found that the adsorption parameters (e.g., the standard free-energy change of adsorption and the "saturation" adsorbed amount) as well as the cmc revealed a pronounced odd/even effect. None of the existing theories can interpret the odd/even effect of the surfactant alkyl chain.

TABLE 1: cmc Values of the Sodium Alkyl Sulfates

	n_C						
	8	9	10	11	12	13	14
[cmc] (mM)	129	62.7	30.7	15.5	8.20	4.19	2.20

Experimental Section

Materials. Sodium alkyl sulfates were prepared from pure *n*-alcohols (Jansen 99+%) by sulfation with chlorosulfonic acid (Merk) at 0−5 °C.[38] The recipe given in ref 38 was slightly modified. The synthesis was performed in a CCl_4 medium, and the reaction mixture was neutralized by sodium carbonate (Aldrich 99.9%). The prepared salts were twice recrystallized from a hot 1:1 benzene and ethanol mixture. Due to the recrystallization procedure, the amount of contaminants in the purified samples is significantly lower than that in the original alcohol used in the synthesis. A combined liquid chromatographic/mass spectrometric analysis showed that the total amount of the other homologues salts is less than 5×10^{-3} mol fraction. The solutions were prepared from dried (for 72 h at 40 °C) surfactants and bidistilled water. The critical micelle formation concentrations (cmc) were determined from the intercept in the specific conductance vs concentration functions (Table 1).

Surface-Tension Measurements. Surfactant solutions always contain impurities, mainly alcohol and longer chain homologues, with surface activities significantly higher than that of the investigated surfactant. The general experience with surfactant solutions shows that bulk purification methods (e.g., recrystallization) do not provide sufficiently pure samples from the surface chemical point of view. Since the highly surface active impurities accumulate in the surface layer of the solution, sufficiently pure samples can be prepared by the repeated removal of the contaminated surface layer of the solution, which can be done either by foaming[39,41] or by the repeated sucking off of the surface.[40] In the present work, surface purification was done by an improved version of the traditional foam fractionation[41] called the gas dispersion method.[30]

Prior to the measurements, gas dispersion was formed in the surfactant solution ($c \approx$ cmc, see Table 1) by producing microbubbles pressing nitrogen through a G-5 sintered glass filter into the solution. The solution was stirred, and after the formation of a stationary foam column, the top of the foam column was continuously sucked off. The mean bubble diameter was about 10 μm, the total generated area was approximately 100−200 m^2/hour, and the purification procedure lasted for a few hours, during which the concentration of the surfactant solution somewhat decreased. The concentration change of the surfactant solution was monitored by means of electrical conductivity measurements.

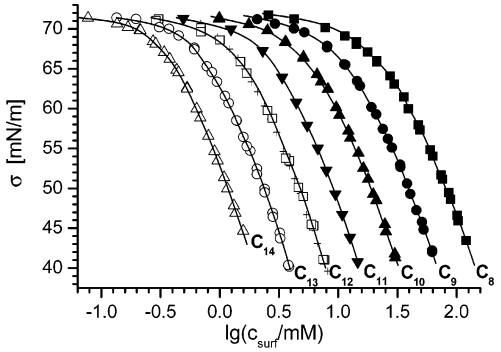


Figure 2. Surface tension plotted against the logarithm of the surfactant concentration for the sodium alkyl sulfate homologues. Surface-tension data measured by Hines for SDS are denoted by +. (The lines are only guides to the eye.)

The surface-tension measurements were performed on surfactant solutions prepared by the dilution of the purified stock solution. Since the alkyl sulfate surfactants are known to hydrolyze slowly to alcohols, the stock solutions were used within a few hours after the purification. To exclude the possibility of contamination caused by the dilution, we made a few test measurements by repeating the purification after the dilution of the surfactant solution. The results of the two kinds of measurements were identical within experimental error, indicating the reliability of the applied method.

The surface tension was calculated from the maximum force measured by the Wilhelmy plate method using a 2.0 cm wide and 0.2 mm thick platinum plate. The purification of the solutions and the dilution were made in a cylindrical cell of 8 cm diameter. The measurements were performed at 25.00 ± 0.05 °C. The concentration of the solutions was determined from electrical conductivity measurements by means of a calibration curve. Five parallel measurements were made at each concentration, and between the consecutive measurements the solution surface was sucked off to form a fresh surface and to eliminate the possible effect of contamination from the environment. The measurements were performed 1−2 min after the fresh surface formed. Time dependence was not observed in the investigated concentration range of the alkyl sulfate solutions. The reproducibility of the maximum force was $\pm 2 \times 10^{-6}$ N (that corresponds to ±0.05 mN/m surface tension).

To check the efficiency of the applied purification method, the surface tension of purified SDS solutions were compared to reference literature values. The measured surface-tension values (53.03 ± 0.05 mN/m at 4 mM and 68.6 ± 0.05 mN/m at 1 mM) were found to be in good agreement with the "best" values given by Mysels[42] (53.0 mN/m at 4 mM and 68.2 mN/m at 1 mM) and Hines[43] (52.8 mN/m at 4 mM and 69.1 mN/m at 1 mM). For comparison, the surface-tension data of Hines is also plotted in Figure 2.

Determination of the Gibbs Surface Excess Concentrations. The adsorption isotherms were calculated from the measured surface-tension data by means of the Gibbs equation

$$\Gamma^{(w)} = -\frac{1}{2RT}\left(\frac{\partial\sigma}{\partial \ln(\gamma_{\pm}c)}\right)_{T,p} \quad (1)$$

where $\Gamma^{(w)}$ is the Gibbs excess relative to the water and $\gamma_{\pm}$ is the mean activity coefficient of the surfactant. $\gamma_{\pm}$ was calculated from the Debye−Huckel limiting law. The activity correction was used a little beyond the accepted range of validity of the Debye−Huckel Law. The derivative was calculated by averaging the slopes of two adjacent data points, resulting in a slight smoothing of the experimental adsorption isotherms. Each series of the parallel measurements was individually evaluated. The standard error of the data points was found to be $s_{\bar{x}} = 7 \times 10^{-12}$ mol/cm^2 from repeated measurements.

It should be noted that the Gibbs excess ($\Gamma^{(w)}$) determined from the surface-tension measurements and the surfactant concentration in the monolayer (Γ_m) used for the interpretation of the experimental data are different quantities. $\Gamma^{(w)}$ consists of two contributions: the monolayer concentration (Γ_m) and the deficiency of the negative surfactant ions in the diffuse electric layer near the negatively charged adsorbed monolayer (Γ_{diff}). However, it can be shown[30] that the two quantities differ from each other significantly only at extreme low surface coverage in the case of the investigated surfactants. In the experimentally investigated concentration range, the contribution of Γ_{diff} to the Gibbs excess is only a few percent. Since this is less than the experimental error, the monolayer concentration can be well-approximated by the Gibbs excess.

Results and Discussion

In Figure 2, the equilibrium surface-tension isotherms of the investigated sodium alkyl sulfates are plotted. The surface-tension isotherms are shifted regularly toward smaller concentrations with increasing alkyl chain length. This observation is in sharp contrast with previous measurements[37] where rather irregular spacing was observed between the surface-tension isotherms of the same homologous series. The observed irregularity was interpreted in terms of an odd/even effect (which is a strong alternation effect of the adsorption parameters observed within the homologous series). Since this alternation effect should also be revealed by parameters like the cmc values and the adsorbed amount at the cmc, we have also checked the effect of alkyl chain length on these parameters.

Due to the purification procedure, the concentration of our surface chemically pure stock solutions was always smaller than the cmc, and we could not determine the adsorbed amount at the cmc directly. To overcome this problem, we plotted the adsorbed amount as a function of relative surfactant concentration (c/cmc) for each surfactant in the vicinity of cmc (Figure 3a). As indicated by the figure, the measured adsorbed amount values are independent of the alkyl chain length at a given relative surfactant concentration within experimental error, which means that odd/even effect cannot be detected.

In Figure 3b, the logarithm of the critical micelle formation concentration (cmc) and that of the surfactant activity at the critical micelle formation concentration (a_{cmc}) are plotted in the function of the alkyl chain length (n_C). The log(cmc) vs n_C function slightly deviates from linearity at small chain lengths, but the log(a_{cmc}) curve is linear in the $n_C = 8-14$ range (with a slope of -0.269 ± 0.001), which also indicates that there is no odd/even effect.

The discrepancy between our results and the previous observation of the odd/even effect may be explained in terms of sample purity. It is well-known that highly surface-active impurities tend to decrease the surface tension of the solution. Consequently, by comparing the surface-tension data of two samples having the same surfactant concentration, their relative purity compared to each other can be determined. In Figure 3c, we plotted the surface-tension data measured for each surfactant at $c = 0.25$cmc. As indicated by the figure, our surface-tension data are considerably higher than the surface-tension values measured in ref 37 in the case of each surfactant, which indicates the presence of impurities in the latter case. However, this seems to contradict the fact that in ref 37 a tedious surface purification

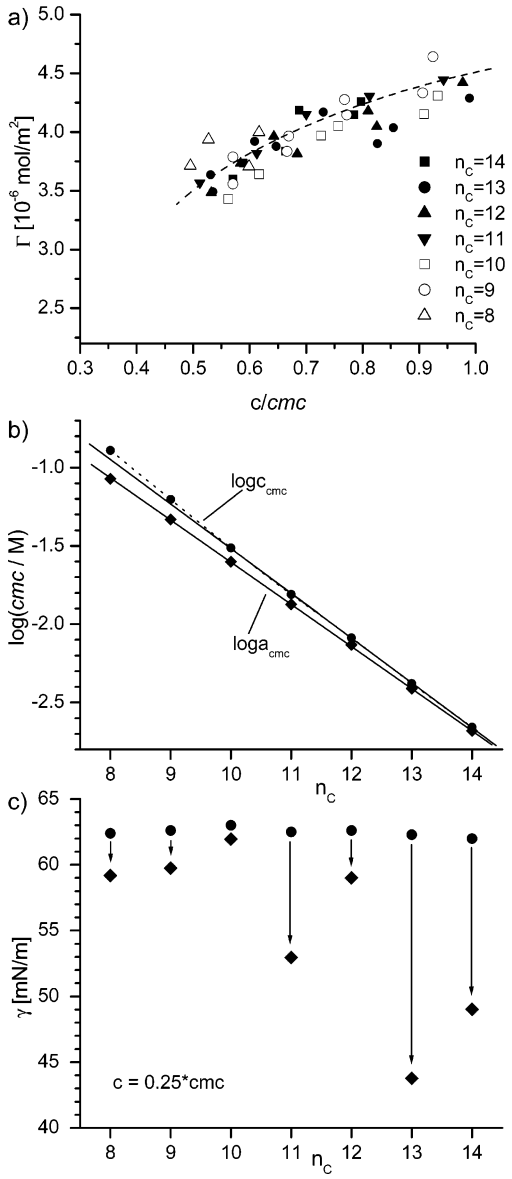


Figure 3. (a) Adsorbed amounts measured near the cmc for the sodium alkyl sulfate homologues. (b) log c_{cmc} and log a_{cmc} as a function of the alkyl chain length n_C. (c) The surface-tension values measured for the sodium alkyl sulfate homologues at 0.25cmc in this work (solid circle) and in ref 37 (solid diamonds).

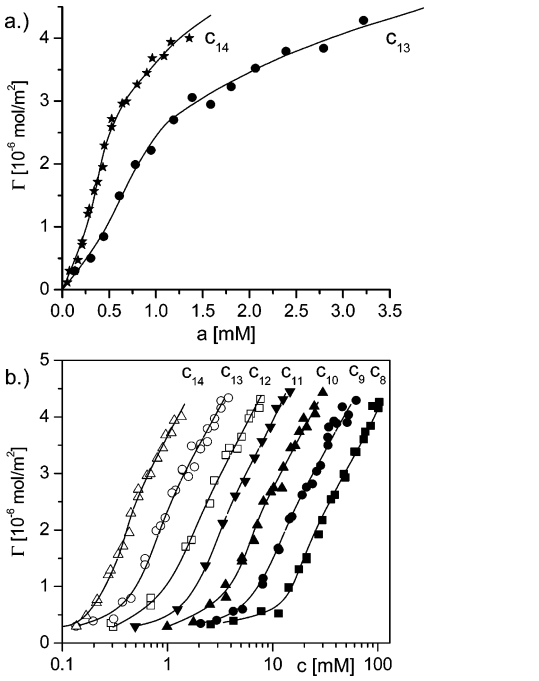


Figure 4. (a) Adsorption isotherms of the sodium tetradecyl and tridecyl sulfate. (b) The adsorption isotherms of the sodium alkyl sulfate homologues are plotted in logarithmic scale. (The lines are only guides to the eye.)

procedure was applied to prepare surface chemically pure solutions. Nevertheless, according to our experience, the surface purification methods can be used only for the removal of trace impurities. If the samples are contaminated in a bulk amount (e.g., if the contaminations exceed 1%), surface purification does not have an effect on the sample purity because only a fraction of the contaminant can be removed by these methods.

In Figure 4a, the adsorption isotherms of sodium tetradecyl and tridecyl sulfate are plotted as an example. The shape of the isotherms is similar in the case of each investigated homologue. The isotherms start with a linear or slightly accelerating region then turn to a saturation-type curve, which does not seem to level off before the cmc. In Figure 4b, the isotherms of the homologues are plotted in a semilogarithmic scale. The highest value of the measured adsorbed amounts (near the cmc, $\sim\Gamma_{cmc}$) falls into the range of $4.0-4.5 \times 10^{-6}$ mol/m^2 independent of the alkyl chain length. It should be noted that if the mean activity coefficient of the surfactant was neglected in the first equation, the calculated adsorbed amounts would underestimate the real values. This effect could give rise to $\sim 0.2 \times 10^{-6}$ mol/m^2 error in the adsorbed amount for the SDS at 7.6 mM surfactant concentration and becomes even more pronounced for the short-chain surfactants due to the larger ionic strength of the solutions (e.g., $\sim 0.4 \times 10^{-6}$ mol/m^2 for the SDS at 30.0 mM).

The surface-tension isotherms of the surfactant solutions or their equivalent adsorption isotherms are usually interpreted in terms of a surface equation of state, which are always based on a priori model assumptions giving rise to several fitting parameters (between two and six). Unfortunately, different models can usually provide similar quality fits to the same experimental data; thus, even if a given model describes the experimental data, there is always the question of whether its success is indeed due to the correctness of the underlying physical assumptions or whether this is due to the fortunate interplay of the fitting parameters. A possible solution to this problem could be a model-independent evaluation of the experimental data.

In bulk thermodynamics, a similar problem arises in the description of real chemical systems. To avoid the application of model- (and system-) dependent equation of states, all real systems are treated as if they were ideal ones and then the deviations from ideality are treated in terms of the interaction of the molecules that is described by the activity coefficient. The major advantage of this approach is that it provides a general framework for the description of the chemical systems and provides a single parameter for the description of nonideality, which is related to molecular interactions. In the remaining part of the paper, we try to apply a similar approach for surfactant adsorption and demonstrate its usefulness.

As the starting point of our treatment, we chose eq 2 for the description of the ideal adsorption layer.

$$\frac{\Gamma}{\Gamma_o - \Gamma} = K(\Gamma)a \qquad (2)$$

Eq 2 is formally identical with the well-known Langmuir isotherm, which describes the adsorption of molecules on N_0 adsorption sites neglecting any interaction among them. In this respect, it should be emphasized that the right-hand side of the equation describes the entropy associated with the occupation of N adsorption sites out of N_0. Any other entropy change of the system (e.g., originating from the conformation difference of the alkyl chains between the bulk and the adsorption layer) and all the interaction of the adsorbed molecules are incorporated in $K(\Gamma)$. $K(\Gamma)$ is an apparent equilibrium constant of the adsorption, which is related to the actual adsorption driving force at any given adsorbed amount ($\Delta G_{tot}(\Gamma)$)

$$\Delta G_{tot}(\Gamma) = -kT \ln(K(\Gamma)) \quad (3)$$

Taking into account that the actual adsorption driving force can be given as

$$\Delta G_{tot}(\Gamma) = \Delta G^\circ + G_{Int}(\Gamma) \quad (4)$$

where ΔG° represents the driving force of surfactant adsorption to the infinitely dilute adsorption layer ($\Gamma \rightarrow 0$) and G_{Int} accounts for the interactions of the adsorbed molecules, the experimentally determinable ($-\ln(K) = \Delta G_{tot}(\Gamma)/kT$) vs Γ function provides direct, model-independent information about the changes of the lateral interaction of the adsorbed surfactant molecules in the function of the surface coverage.

To determine $K(\Gamma)$ as the function of the adsorbed amount, one has to choose an appropriate value for the saturated adsorbed amount (Γ_o, see eq 2). Unfortunately, the saturated adsorbed amount values, used in the literature for the interpretation of sodium alkyl sulfate surface-tension data, scatter in an extremely wide range ($\sim 3.5-8.0 \times 10^{-6}$ mol/m^2). With a closer look at the data, it can be concluded that the literature values of Γ_o fall into two groups. When they are determined by fitting surface-tension data with an equation of state, then the fitted values fall into the range of $\sim 3.5-5.0 \times 10^{-6}$ mol/m^2.[2] On the other hand, when Γ_o is determined or estimated from an independent source, it is in the range of $\sim 6-8 \times 10^{-6}$ mol/m^2.[28-30,48] The reason of the large discrepancy between the Γ_o values provided by the two approaches basically lies in the characteristic features of parameter fitting.

To get a good quality estimate for the limiting value of a saturation-type function by parameter fitting, experimental data are necessary from the saturation region of the curve. If experimental data are available only in the increasing part of the function (as seems to be the case for the sodium alkyl sulfate adsorption isotherms), the limiting value becomes highly correlated with the rest of the parameters. In practice, this means that similar quality fits can be achieved to the experimental data by fixing the saturation value at practically any value in an unacceptably wide range, thus making the fitted parameter values unreliable. As a consequence, in such a case the saturated adsorbed amount must be determined by an independent method.

According to eq 2, in the case of surfactant adsorption the adsorbed amount tends to saturation if $K \cdot x_{surf} \gg 1$, where x_{surf} is the mole fraction of the surfactant monomer in the solution. However, since the micelle formation places a cap on the chemical potential and thus on the concentration of the surfactant, the value of Γ_{cmc} can be much smaller than that of Γ_0. If this is the case for sodium alkyl sulfate surfactants, one can compare the experimentally determined adsorbed amounts to values measured for surfactants, which have a similar structure (single-tailed, small-head groups) but achieve higher $K \cdot x_{surf}$ values in the vicinity of their cmc. In the case of nonionic and catanionic surfactants, where the electrostatic contribution does not act against the adsorption, the K values are expected to be several magnitudes larger than in the case of ionic surfactants and the adsorbed amount is expected to approach Γ_o much better. Indeed, alcohols and catanionic surfactants (e.g., alkyl sulfate alkyl trimetylammonium salts) give much larger adsorbed amounts at their solubility limits than sodium alkyl sulfates. In the case of saturated normal alkyl chain alcohols, Γ_o was found to be 7.75×10^{-6} mol/m^2 independent of the alkyl chain length (21.5 Å^2 surface area per molecules) by high-resolution synchrotron X-ray measurements,[44] while the highest measured value of Γ near the solubility limit of the catanionic surfactants[45] was $\sim 6 \times 10^{-6}$ mol/m^2.

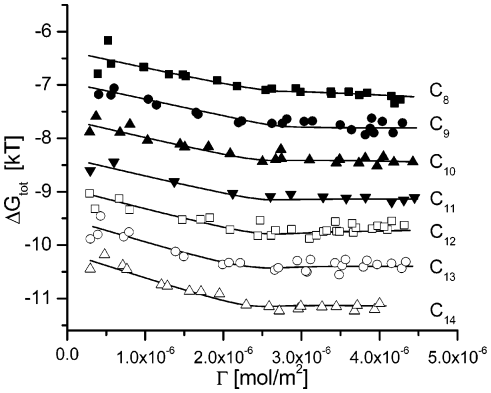


Figure 5. Total adsorption driving force plotted against the adsorbed amount for sodium alkyl sulfates homologues.

Independent estimates for the limiting adsorption of sodium alkyl sulfate surfactants have been derived from different sources in the literature. Warszynski et al.[28,29] used a 26 Å^2 limiting area per molecule, which originates from the analysis of SDS crystal structure data.[46,47] Shiloach and Blankschtein[48] calculated the head area of alkyl sulfate surfactants from bond angle and length data to be 25 Å^2. Recently, Shapovalov et el.[49] investigated the surface pressure area isotherms of insoluble (n_C = 20 and 22) alkali alkyl sulfates. They found that the maximum packing density is $\sim$25 Å^2. Since the data originating from different sources are in very good agreement with each other, we simple used their average in our calculations ($\Gamma_o = 6.6 \times 10^{-6}$ mol/m^2). Finally, it should be noted that even as much as 10% error in the applied Γ_o value would give rise to only $\sim 0.25kT$ error in the calculated driving-force values in the vicinity of the cmc, which is comparable to the statistical error of the results.

The calculated adsorption driving force in the function of the adsorbed amount (ΔG_{tot} vs Γ functions) is plotted for the investigated alkyl sulfate surfactants in Figure 5. As indicated by the figure, the driving force of the adsorption first slightly increases (becomes more negative) with increasing surface coverage then becomes approximately constant with further surfactant adsorption in the case of each homologue. The curves shift regularly with increasing chain length to more negative values. The initial increase of the driving force can be explained qualitatively in terms of the lateral interactions between the alkyl chains of the adsorbed surfactant molecules, since the electrostatic term is expected to give an opposite contribution because it is positive and increases with Γ. On the other hand, the leveling off of the driving force at larger adsorbed amounts may originate either from the electrostatic interaction of the surfactant molecules or from the leveling off of the hydrophobic driving-force contribution. To gain further insight into the characteristics of the alkyl sulfate adsorption, the total adsorption driving force should be divided into an electric and nonelectric contribution

$$\Delta G_{tot}(\Gamma) = \Delta G^\circ + G_{Int}(\Gamma) = \Delta G^\circ + f(\Gamma) + y_0(\Gamma) \quad (5)$$

where $f(\Gamma)$ represents the nonelectrostatic interactions of the adsorbed surfactant molecules and $y_0 = ez\Psi_0/kT$ is the electrical free-energy contribution of the driving force, in which Ψ_0 is the electrostatic potential at the plane of the ionic head groups of the adsorbed surfactant molecules with a valence z. In model calculations, this division is done by the a priori choice of a hydrophobic driving-force function. Traditionally, the Frumkin isotherm is usually used for this purpose. However, the theoretical derivation of the Frumkin isotherm is based on the nearest-neighbor interaction of disklike molecules,[4] so its validity (and consequently, the linear dependence of the hydrophobic driving force on the adsorbed amount) is expected to break down in a surface coverage range where the surfactant alkyl chains start to overlap with each other ($\sim 1 \times 10^{-6}$ mol/m^2). As a consequence, the question naturally arises how the hydrophobic driving force can be described in the function of the adsorbed amount.

A possible approach to get information about the hydrophobic driving force could be the calculation of the electrostatic contribution of the driving force (y_0) and its subtraction from the experimentally determined total driving force

$$\Delta G_{Hydrofob}(\Gamma) = \Delta G^\circ + f(\Gamma) = \Delta G_{tot}(\Gamma) - y_0(\Gamma) \quad (6)$$

Unfortunately, the calculated y_0 values are model-dependent and even within the framework of a given electrostatic model not only a unique set of parameters can be used for the description of the experimental data. Practically, by scanning the physically meaningful region of the parameter space of the electrostatic model, one can determine a $\Delta G_{Hydrofob}$ vs Γ function for each set of the electrostatic parameters. It should be emphasized that since the sum of a corresponding pair of electrostatic and hydrophobic driving-force functions is equal to the experimentally determined total driving-force function, each pair of the determined driving-force contributions gives a perfect fit to the experimental surface-tension data. Thus, the most important information provided by such an analysis is not the exact value of the model parameters but the functional forms of the hydrophobic driving-force function compatible with a given electrostatic model.

To investigate what type of hydrophobic driving-force functions can provide consistent results with current ionic surfactant adsorption theories, we performed the above analysis for the models of Kalinin and Radke,[21] Warszynski et al.,[29] and Gilanyi et al.[30] The relevant details of the calculations can be found in the original publications; here, we only summarize the electrostatic parameters introduced into the different models and their effect on the calculated hydrophobic driving-force functions.

The model of Kalinin and Radke[21] contains three different electrostatic parameters: The equilibrium constant characteristic to the counterion binding ($RSO_4^- + Na^+ \overset{K}{\leftrightarrow} RSO_4Na$) and the integral capacitances of the inner ($C_{0\beta}$) and outer ($C_{\beta\delta}$) Stern layers (see Figure 1b). In the original publication, the two capacitance values were set a priori to "reasonable" values and the third parameter was fitted. The analysis of the electrostatic model indicates that the parameters can be divided into two groups. On one hand, the capacitance of the inner Stern layer has a profound effect on the shape of the calculated hydrophobic function (see Figure 6a). If the internal Stern-layer is assumed to have a small capacitance, which corresponds to large layer thickness ($C_{0\beta} = \epsilon_0 \cdot \epsilon_{r,0\beta}/\delta_{0\beta}$, where ϵ_0 is the vacuum permittivity,

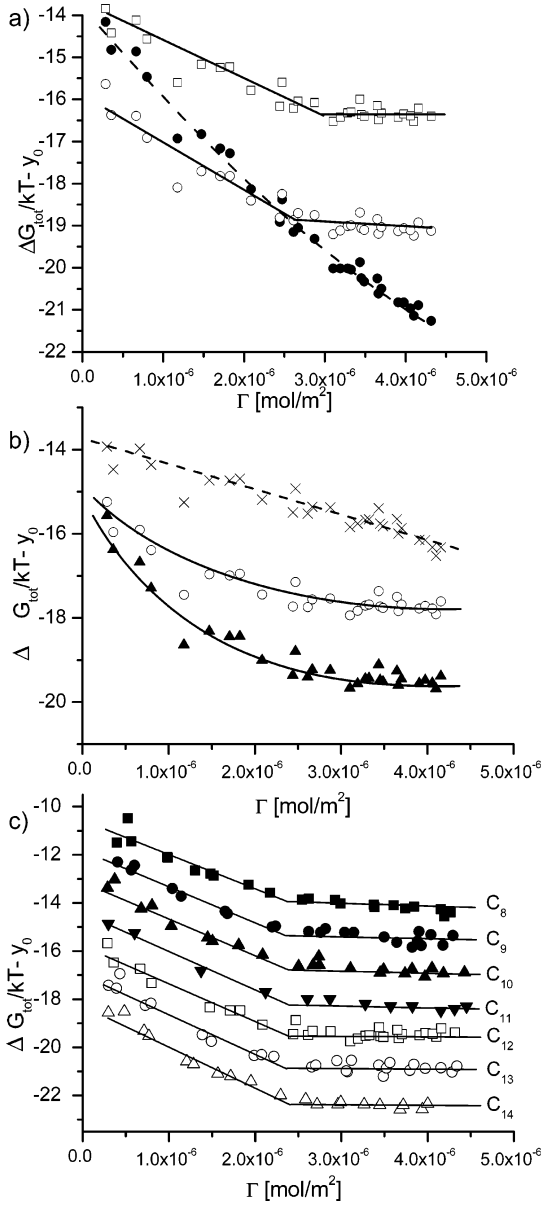


Figure 6. Calculated hydrophobic driving force of adsorption in the function of the adsorbed amount. (a) Determined for SDS by means of the model of Kalinin and Radke with parameters K (dm^3/mol), $C_{0\beta}$ (C^2/J·m^2), and $C_{\beta\delta}$ (C^2/J·m^2). Key: ◆ (61.3, 2.0, 7.0); □ (61.3, 5.0, 7.0); ○ (1.0, 5.0, 7.0). (b) For SDS calculated by means of the model of Warszynski et al. with parameters K (dm^3/mol), $C_{0\delta}$ (C^2/J·m^2), ϵ_r, g_{sc}, r_{ef} (Å). Key: × (0.005, 2.6, 30.0, 0.6, 3.7); ○ (0.005, 6.9, 78.4, 0.5, 3.7); ▲ (0.001, 3.5, 78.4, 0.5, 3.7). (c) Calculated for the sodium alkyl sulfates homologues by means of the model of Gilanyi et al.

$\epsilon_{r,0\beta}$ is the dielectric constant, and $\delta_{0\beta}$ is the thickness of the inner Stern layer), the predicted hydrophobic driving-force function is almost linear. However, with increasing capacitance (decreasing layer thickness) the deviation of $\Delta G_{Hydrofob}$ vs Γ function from linearity increases and it becomes a saturation-type curve. On the other hand, by changing the equilibrium constant of the counterion binding or the capacitance of the outer Stern layer, the hydrophobic driving-force function merely shifts without changing its functional form (see Figure 6a).

The model of Warszynski et al.[29] (see Figure 1c) contains five different electrostatic parameters, the equilibrium constant of the counterion binding ($M + Na^+ \overset{K}{\leftrightarrow} MNa^+$, where M denotes an adsorption site at the interface), the effective counterion radius in the adsorption layer (r_{ef}), the ratio of surfactant headgroup size to the adsorbed counterion size (g_{sc}),

the capacitance ($C_{0\delta}$), and the dielectric constant of the Stern layer (ϵ_r). The shape of the calculated hydrophobic driving-force function can resemble both linear and saturation-type functions as depicted in Figure 6b. Although the actual shape of the predicted $\Delta G_{\text{Hydrofob}}$ vs Γ function is determined by a complex interplay of the electrostatic parameters, two of the five parameters (the capacitance of the Stern layer and the ratio of surfactant headgroup size to the adsorbed counterion size) have the most significant effect. When these parameters adopt a value that favors a smaller amount of counterion binding (large capacitance and large counterion size), then the hydrophobic driving-force function becomes a saturation-type curve. By changing the rest of the parameters, the hydrophobic driving-force function is shifted above all.

The counterion penetration model of Gilanyi et al.[30] does not contain any electrostatic parameters for counterions that are smaller than the surfactant headgroups and contain one parameter, the capacitance of the Stern layer, for counterions that are larger than the surfactant headgroups. However, since in this model the capacitance is uniquely determined by the type of the counterion and the surfactant headgroup (e.g., $C = 15.27 \pm 0.78$ C^2/J·m^2 for Na^+ and dodecyl sulfate), the shape of the hydrophobic driving-force function is also uniquely determined in the framework of this model. As can be seen in Figure 6c, this model gives rise to saturation-type $\Delta G_{\text{Hydrofob}}$ vs Γ functions.

To summarize, it can be concluded that two types of hydrophobic driving-force functions seem to be compatible with current electrostatic theories. On one hand, if it is assumed that the adsorption layer of sodium alkyl sulfates can be characterized by a small Stern-layer capacitance (in other words, there is a thick, charge-free Stern layer between the adsorption plane and the Helmholtz plane, $\delta \approx r_{\text{head}} + r_{\text{Counterion}}$), then the experimental adsorption data allow a linear (Frumkin-type) hydrophobic driving-force function. On the other hand, if it is assumed that the Stern-layer capacitance is large ($\delta < r_{\text{head}} + r_{\text{Counterion}}$), then the hydrophobic driving-force function reaches a plateau value and becomes constant independent of the applied electrostatic model.

Unfortunately, considering only the adsorption isotherms, it is not possible to decide which of the above two cases provides a more realistic description of the surface-coverage dependence of the hydrophobic driving force. However, the results of recent neutron-reflection, sum-frequency generation spectroscopy, and ellipsometry measurements as well as the results of Monte Carlo simulations seem to provide support for the saturation-type hydrophobic driving-force function. On one hand, it was concluded from neutron-reflection measurements[34] made on decyl trimethylammonium bromide solutions that the alkyl chains probably form a liquidlike layer with large-amplitude fluctuations in their distribution normal to the surface. On the other hand, a combined sum-frequency spectroscopy and ellipsometry study[35] also found that the monolayer has a density close to that of a liquid hydrocarbon in the case of alkyltrimethylammonium bromides. Finally, a recent Monte Carlo investigation of octanol adsorption at the vapor/solution interface showed that above $\Gamma = 2 \times 10^{-6}$ mol/m^2 surface excess concentration the density of the adsorption layer becomes approximately constant and only its thickness increases with additional octanol adsorption.[33] All of these results imply that the adsorbed molecules form a gaslike adsorption layer only at low surface coverage. In this low surface coverage range, the surfactant molecules experience increasing lateral interactions as the surface coverage increases. However, when the surface coverage becomes large enough, the overlapping alkyl chains of the adsorbed surfactant molecules start to resemble a thin liquidlike alkane layer. With further increasing adsorbed amount, the thickness of this layer increases, but the local environment and thus the interactions of the alkyl chains remain practically the same, resulting in the leveling off of the hydrophobic driving-force function.

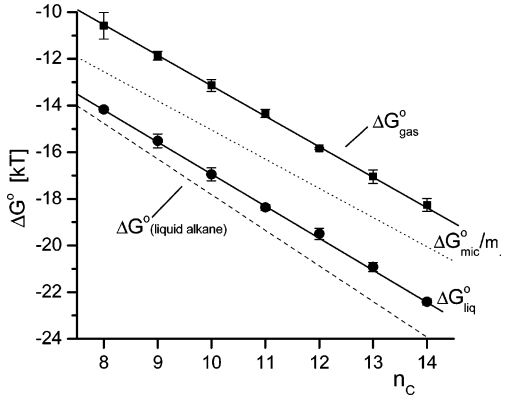


Figure 7. The free-energy change as a function of the alkyl chain length n_C, with $\Delta G°_{\text{gas}}$ and $\Delta G°_{\text{liq}}$ from the adsorption measurements, dashed line $\Delta G°$(liquid alkane) from the solubility of liquid alkanes, and dotted line $\Delta G°_{\text{mic}}/m$ from the hydrophobic contribution for the micelle formation.

Assuming that the molecular interactions that govern the hydrophobic driving force at larger surface coverage indeed resemble the interactions characteristic for a liquid alkane environment, it is interesting to determine the alkyl-chain-length dependence of the hydrophobic driving-force functions and compare them to alkane solubility data. For this purpose, we used the counterion penetration model of Gilanyi et al., since it does not contain any adjustable parameters. However, it should be noted that the other two counterion binding theories give similar results with appropriately chosen parameters. In Figure 6c, the nonelectrostatic part of the adsorption driving force ($\Delta G_{\text{Hydrofob}}$) is plotted for the alkyl sulfate homologues. The calculated data reach a plateau value and become constant independent of the alkyl chain length and seem to shift regularly toward larger values with increasing alkyl chain length.

As a first approximation, the hydrophobic driving-force data can be divided into two regions, an increasing part and a roughly constant part. By extrapolating the data of the first regions of the hydrophobic driving-force functions (Figure 6c) to zero adsorbed amount, the classical standard state free-energy change of adsorption can be derived, which is characteristic for the adsorption of an individual surfactant without lateral interactions. The tangent of the same region provides the pair interaction parameter of the surfactant molecules that could be used for the characterization of the lateral interaction of the adsorbed molecules before the alkyl chains start to overlap. On the other hand, the limiting value of the second region of the hydrophobic driving-force function is characteristic for the transfer of the alkyl chains from the aqueous to a liquidlike alkane environment. For the sake of simplicity, the two standard energies will be distinguished as $\Delta G°_{\text{gas}}$ and $\Delta G°_{\text{liq}}$. In Figure 7, the determined $\Delta G°_{\text{gas}}$ and $\Delta G°_{\text{liq}}$ values are plotted as a function of the alkyl chain length (n_C). As indicated by the figure, both energy functions change linearly with n_C

$$\Delta G°_{\text{liq}}/kT = -3.23 - 1.37n_C \quad (7a)$$

$$\Delta G°_{\text{gas}}/kT = 0.03 - 1.32n_C \quad (7b)$$

The plausibility of the determined standard free energies can be checked by comparing them to standard free-energy changes determined from alkane solubility data. The free-energy change calculated from the solubility of liquid alkanes in water[50] can be given in the following form: $\Delta G°(\text{alkane})/kT = (-4.131 - 1.527n_C)$ (see the dashed line in Figure 7 below $\Delta G°_{liq}$). On the other hand, in the molecular interaction models of the surfactant micelle formation the $\Delta G°_{mic}/mkT = (-2.55 - 1.25c_n)$ relation is used for the hydrophobic driving force of the micelle formation,[51] which is derived from the alkane solubility data with a semiempirical correction for the entropy loss of the alkyl chains in the micelle core. This function is also plotted in Figure 7 (see the dotted line above $\Delta G°_{liq}$). Taking into account that the conformational freedom of the alkyl chains confined to the thin adsorbed layer should be smaller than that of the alkyl chains found in a liquid alkane phase but should be larger than that of an alkyl chain confined to a micelle, the hydrophobic driving-force function determined for the transfer of the alkyl chains from the aqueous phase to the liquidlike adsorption layer seems to be reasonable.

The standard state free energy of adsorption ($\Gamma \rightarrow 0$) can be given as: $\Delta G°_{gas} = -0.88 - 1.22n_C$. This relation significantly differs from the one calculated from the solubility of the standard state (25 °C and 1 atm gas pressure) alkane gases in water,[50] which is $\Delta G°(\text{aq} \rightarrow \text{gas})/kT = (-9.835 - 0.305\, n_C)$. However, the absolute values of these free energies cannot be compared with each other because the free-energy difference of the standard gas state and the standard state of the adsorbed layer is undefined. However, it is interesting to note that the increment of the driving force with increasing chain length is 4 times larger in the case of the adsorption layer than in the case of the gas phase. This can be explained by the fact that in addition to the energy change of transfer of the alkyl chains from the aqueous to the air phase the alkyl chains can interact with the water surface in the adsorbed layer.

Conclusions

We performed surface-tension measurements on surface chemically pure sodium alkyl sulfates ($n_C = 8-14$). The measured surface-tension isotherms and cmc values changed regularly, and no sign of the so-called odd/even effect could be observed. The discrepancy with an earlier investigation was interpreted in terms of the purity of the surfactants. We also presented a model-independent approach for the analysis of surface-tension isotherms, which is based on the calculation of the model-independent total adsorption driving-force data in the function of the adsorbed amount (surface coverage). Our analysis indicated that current ionic adsorption models seem to be compatible with both linear and saturation-type $\Delta G_{Hydrofotb}$ vs Γ functions. Although independent experimental results (neutron reflection, ellipsometry, sum-frequency generation spectroscopy) as well as alkane solubility data seem to support the existence of saturation-type hydrophobic driving-force functions, the clarification of the surface-coverage dependence of the hydrophobic driving force requires further experimental and theoretical investigations.

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